Radical majorants, guard compression, and Pell descent for Diophantine consistency equations

A research contribution prepared for Vladimir Reshetnikov's ProveIt project

Mathematical development and executable audits with AI assistance

23 October 2026

Abstract

We construct four explicit integer polynomials whose solvability is equivalent to the existence of a finite ZFC refutation, by composing the repository's ordinary native calibration with new, exactly verified arithmetic reductions. The resulting unrestricted-integer forms have variable/degree pairs $(31, 6)$, $(26, 8)$, $(23, 12)$, and $(22, 14)$. In particular the last form replaces the inspected $(26, 16)$ frontier by four fewer variables and a lower degree. A direct polynomial-part majorant also lowers the degree of the nine-positive-variable construction from 1,379,730 to 1,158,228, with a larger circuit. A separate positive-integer construction preserves eleven variables and reduces degree 325 to 277, using 111 live binary arithmetic operations. The central integer idea combines a canonically calibrated determinant inequality with a universal simultaneous-positivity identity, then chooses coordinate elimination and polynomial row operations jointly. The positive construction removes an unnecessary denominator guard by a rational Pell descent. All coefficient tuples are fixed and reproduced exactly. The new algebraic reductions and degree certificates are checked independently. Their consistency interpretation uses the inherited ordinary native/Pell calibration and its canonical completeness theorem. This work does not independently re-establish that complete semantic bridge or supply its end-to-end Lean/Rocq verification or an internally encoded ZFC derivation.

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1 The target and the comparison

Fix the recursive syntactic proof predicate used by the repository and let Con(ZFC) mean that it has no finite proof of contradiction. The target for a polynomial $P \in \mathbb{Z}[x_1, \ldots, x_n]$ is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con(ZFC)}. \tag{1}$$

For the nine- and eleven-variable constructions, replace $\mathbb{Z}^n$ by $\mathbb{Z}^9_{>0}$ and $\mathbb{Z}^{11}_{>0}$, respectively. Neither this statement nor its proof asserts the truth of Con(ZFC).

The inspected repository state is the full commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

It includes all wave-ten packages 68–75. Its active review records the frontier in Table 1; the table below compares domains and degrees explicitly. There is no established global minimum for the user's word "simplest." A lower variable count, lower degree, shorter arithmetic circuit, and smaller coefficients are distinct objectives.

Domain / criterion	Inspected frontier	New construction	Residuals
Integers, degree 6	32 variables	**31 variables**	18
Integers, degree 8	27 variables	**26 variables**	14
Integers, degree 12	no smaller-degree counterpart	**23 variables**	11
Integers, degree at most 16	26 variables, degree 16	**22 variables, degree 14**	10
Positive integers, 11 variables	degree 325, 113 gates	**degree 277, 111 gates**	2
Positive integers, 9 variables	degree 1,379,730	**degree 1,158,228**	1

Table 1: New explicit tradeoffs against the pinned wave-ten review. Each integer polynomial is the sum of the squares of its listed residuals. The eleven-positive equation uses a product finalizer; the nine-positive equation uses a two-root norm. No optimized scalar-circuit claim is made for the integer forms.

The new $(23, 12)$ form is smaller in both variables and degree than the old $(26, 16)$ form. It trades degree against the old $(27, 8)$ form. The new $(22, 14)$ form improves that older $(26, 16)$ point by $4/26$, or about 15.38%, in variable count, while lowering degree from 16 to 14. These improvements do not lower the already available nine-positive or ten-unrestricted-integer variable bounds.

What is new here. The determinant inequality is from package 71, the simultaneous-positivity lemma from package 68, the signed-scale bootstrap from package 75, and the rational descent from package 69. We do not claim priority for these ingredients. The new results are their compatible composition, the resulting literal equations and elimination schedules, and the observation that reducing an auxiliary norm modulo a retained strong norm avoids an unnecessary degree increase; centering the remaining root then reduces it further. The independent polynomial-part majorant is a different simplification that retains nine positive variables and reduces degree by about 16.05%. None of these combinations is in the pinned wave-ten frontier. No historical novelty outside the inspected sources is asserted.

2 A complete 23-integer equation

All 23 variables in the following ordered tuple range independently over *all* integers:

$$(J, F, \alpha, f, h, r, v, \tau, \eta, y, \gamma, o, j, E, c, a, I, u, Z, t, p_1, p_2, p_3).$$

The six fixed integer coefficients are

$$\beta, K_0, M_C, M_F, A_* = 2^{1113},\ B_* = 2^{666104}.$$

Here $\beta = 2^{1064921} - 1$, and the first four integers are exactly the whole source-34 tuple specified in Section 7. In particular M_F is already shifted. None of these six integers is an unknown.

Define the following *polynomial abbreviations*; they introduce no quantifiers or extra equations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & Y &= (2v+1)q^3,\\
X &= 4qu^2, & H &= 4a + 3, & \Delta &= a^2 + H,\\
V &= fo - c, & k &= 1 - V + jc + hE, & \zeta &= k - \eta,\\
P &= k + E(c - \eta), & D &= X + ac + \gamma H. &&
\end{aligned} \tag{2}$$

The packing and sign registers are

$$\begin{aligned}
R &= ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,\\
U_* &= A_* F, \qquad W_* = B_* C, \qquad G = U_* W_* - (q - U_* - W_*)^2,\\
e &= \eta\zeta, \qquad \mathrm{Pos}(p_1, p_2, p_3) = 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{aligned}$$

The eleven residuals are

$$\begin{aligned}
b_1 &= a - E - Y, & b_2 &= EX + E - Xa,\\
b_3 &= c - (a - E)k - \eta, & b_4 &= \tau^2 - P^2 + Pk - 1,\\
b_5 &= D^2 - \Delta c^2 - 1, & b_6 &= jc - V - R,\\
b_7 &= (K_0 q + X)C - qF - Z(q-1), & &\\
b_8 &= I - (2r+1)c^2, & b_9 &= I^2 - \Delta(f^2 - 1),\\
b_{10} &= y^2 - I^2(y^2 - V^2) - 1, & &\\
b_{11} &= Ge - (t^2 - G - e + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3). & &
\end{aligned} \tag{3}$$

The promised single ordinary polynomial is

$$\boxed{P_{23} = \sum_{i=1}^{11} b_i^2 = 0.} \tag{4}$$

Every exponent is fixed and nonnegative. There are no divisions, variable exponents, bounded universal quantifiers, or unnamed auxiliary witnesses in this equation. The variable called `v_scale` in the executable packet is v above.

Theorem 2.1. *The polynomial P_{23} has exact total degree 12 after the six fixed coefficients are specialized to their actual values. Composing the ordinary source-34 native calibration with the reductions proved below yields (1) for $P = P_{23}$.*

The exact degree assertion does not depend on a generic-coefficient argument. The coefficient of $\eta^4 f^4 o^4$ in P_{23} is exactly 1. Indeed the degree-three part of e contains $-\eta f o$; the only residual that can contribute this monomial after squaring is b_{11}, whose relevant term is $-\eta^2 f^2 o^2$. Every residual has degree at most six. The independent exact expansion checks this coefficient and all eleven compact formulas against the separate elimination construction.

3 Compressing the sign constraints

3.1 One universal equation for two positive integers

For unrestricted integers x, y, t, put

$$\Gamma(x, y, t) = xy - (t^2 - x - y + 1)^2.$$

Lemma 3.1 (Nearest-square positivity)**.** *For integers x, y,*

$$x > 0 \ \wedge \ y > 0 \quad \Longleftrightarrow \quad \exists t \in \mathbb{Z}\ \Gamma(x, y, t) > 0.$$

Proof. If $\Gamma > 0$, then $xy > 0$, so x, y have the same nonzero sign. If both were negative, say $x = -a, y = -b$ with $a, b \geq 1$, then $(t^2 + a + b + 1)^2 > ab = xy$, a contradiction.

Conversely put $N = x + y - 1 \geq 1$. Then $xy - N = (x-1)(y-1) \geq 0$. Choose the square nearest to N as follows. With $m = \lfloor \sqrt{N} \rfloor$, take $t = m$ if $N - m^2 \leq m$, and $t = m + 1$ otherwise. If N is a square, the error is zero. Otherwise $|t^2 - N| \leq m$ and $m^2 < N$. In either case $(t^2 - N)^2 < N \leq xy$, proving the claim. □

Every value of Pos is a positive integer. Conversely every positive integer n is a value of Pos: apply the three-square theorem to $4n - 3$, which is 1 or 5 modulo eight. Exactly one square is odd and two are even, giving $4n - 3 = (2p_3 + 1)^2 + (2p_1)^2 + (2p_2)^2$. Therefore

$$x, y > 0 \quad \Longleftrightarrow \quad \exists t, p_1, p_2, p_3 \in \mathbb{Z}\ xy - (t^2 - x - y + 1)^2 = \mathrm{Pos}(p_1, p_2, p_3). \tag{5}$$

This replaces six private variables in two separate Pos equations by four variables in one equation. It is an existence equivalence, not an identity of the old and new witness sets.

3.2 A determinant with a known canonical orientation

The source Q41 has the three positive-product conditions

$$F\alpha > 0, \qquad FC > 0, \qquad \eta\zeta > 0, \qquad C = q - F - \alpha.$$

Package 71 replaces the first two with $G > 0$, for the G in (2). The following universal part is short enough to repeat.

Lemma 3.2 (Determinant soundness)**.** *For arbitrary real F, C, α, q with $q = F + C + \alpha$ and fixed $A_*, B_* \geq 2$, the inequality $G > 0$ implies $F\alpha > 0$ and $FC > 0$.*

Proof. Write $U = A_* F$, $W = B_* C$, and $T = q - U - W$. The inequality $UW > T^2$ makes U, W nonzero with the same sign, and $|T| < \sqrt{UW} \leq (|U| + |W|)/2$. Consequently $q = U + W + T$ has their sign and

$$|q| > (|U| + |W|)/2 \geq |F| + |C|.$$

It follows that $\alpha = q - F - C$ has this same nonzero sign. The argument includes negative q and requires no native or Pell assumption. □

Completeness cannot be inferred for arbitrary old positive triples. It uses the source's canonical witnesses. For the authentic source-34 tuple, package 71 proves uniformly over every canonical accepting word that

$$q/2 \le U_* < 4q/5, \qquad q/2 \le W_* < 3q/5, \qquad q > 0. \tag{6}$$

For clarity, put $s = q/2$, $U_* = s + x$, $W_* = s + y$. Then $0 \le x < 3s/5$, $0 \le y < s/5$ and

$$G = s^2 + s(x + y) - (x^2 + xy + y^2) > \frac{12}{25}s^2 > 0.$$

The canonical interval estimates are justified by the fixed leading native digits; Section 7 records the numerical alignment. They survive all rank-tuning dummy changes allowed by that source.

3.3 The new combined guard

Apply (5) with $x = G$ and $y = \eta\zeta$. The single residual $b_{11} = 0$ is equivalent to

$$G > 0 \quad \text{and} \quad \eta\zeta > 0.$$

Thus every new zero restores all three old sign conditions, without any prior sign assumption on a source coordinate. Conversely a canonical source zero satisfies $G > 0$ and $\eta\zeta > 0$, so the nearest-square choice of t and a three-square representation supply the four new coordinates. The original nine private sign coordinates have been replaced by four. This is the main variable-count saving.

4 Why the signed-scale and determinant reductions compose

The Q41 source contains

$$X = 4qw, \qquad X = 2u_{\text{old}}^2, \qquad (K_0 + 4w)C = F + z(q - 1).$$

Package 75 proves that one may replace them by

$$X = 4qu^2, \qquad (K_0 q + X)C - qF = Z(q - 1), \tag{7}$$

removing one scale variable. Its proof first treats both possible signs of X and q. It does not infer $X > 0$ from $X = 4qu^2$.

The inherited signed-bootstrap theorem has these exact two parts:

1. With the other Q41 arithmetic equations and the three old positive product conditions, every integer solution of (7) recovers $q > 0$, $X = 2^R$, and odd R by Pell rank and Mersenne-congruence arguments. It restores $w = u^2$, $u_{\text{old}} = 2^{(R-1)/2}$ and $z = Z/q$. The last quotient is integral by reducing the cleared transport modulo q.

2. Canonical refutation witnesses have $q = 2^s$ with odd s, odd R, and $R > 3q$. The assignments $u = 2^{(R-s-2)/2}$ and $Z = qz$ establish (7); all of $q, F, \alpha, C, \eta, \zeta$ remain unchanged.

The new combined guard supplies the old product conditions before invoking the soundness theorem. In the reverse direction the signed substitution does not change the coordinates in the canonical determinant estimates (6). Thus there is a single compatible canonical witness for both constructions. Equiconsistency alone would not establish this compatibility; the actual coordinate maps do.

For reference, the nontrivial sign argument in package 75 obtains $|X| \geq 4|q|$, $|E| \geq 4|q|^4 > 3R+1$ before fixing a sign. If $X < 0$, Pell growth and the signed rank equality force $|X| = 2^R$ and hence $q = -2^s$. But $q \equiv 1 \pmod{2^{1064921} - 1}$ forbids a negative power of two: reduce s modulo 1064921. This excludes that branch. The complete ordinary proof and its previous independent review are included as inherited sources, rather than replacing the argument by sample calculations.

4.1 Exact elimination

After signed-scale replacement and the combined guard there are 35 integer variables and 23 residuals. The following twelve monic definitions are eliminated successively:

$$\begin{aligned}
&N = q(L+1), && C_2 = c^2, && F_2 = f^2, && g = y^2 - V^2,\\
&X = 4qu^2, && B_{\mathrm{aux}} = I^2, && D = 4qu^2 + ac + \gamma(4a+3), &&\\
&\Delta = a^2 + 4a + 3, && P = \eta + \zeta + E(c - \eta), &&&&\\
&L = q^2 - 1, && \zeta = 1 - V + jc + hE - \eta, && V = fo - c. &&
\end{aligned}$$

Substitution in every remaining row gives exactly (3). Monic elimination over $\mathbb{Z}$ is reversible: the eliminated coordinates have the unique displayed integer values. The archive contains a full parent-row restoration certificate, including the twelve rows that become identically zero. Therefore this last step is a genuine bijection of zero sets, unlike the earlier sign-guard changes.

Combining the all-input soundness, canonical completeness, and exact eliminations proves Theorem 2.1 using the named ordinary source calibration. All newly supplied steps are elementary arithmetic and finite polynomial reasoning available in ZFC.

5 A 22-integer equation of degree 14

An indiscriminate substitution of $I = (2r+1)c^2$ in P_{23} raises the auxiliary residual to degree ten and the single polynomial to degree twenty. There is a better order of operations. Put $W = y^2 - V^2$ and replace the auxiliary residual by

$$b'_{10} = b_{10} + Wb_9 = y^2 - \Delta(f^2 - 1)(y^2 - V^2) - 1. \tag{8}$$

Keeping b_9, this is an invertible polynomial row operation: subtract Wb_9 to recover b_{10}. It therefore preserves the common zero set over every commutative ring, not only over a finite collection of tests.

Now delete I and b_8, substitute $I = (2r+1)c^2$ only in b_9, and use (8). Finally set $y = V + y_t$, where y_t is a new unrestricted integer replacing y. This polynomial shear has inverse $y_t = y - fo + c$ and changes $y^2 - V^2$ to $y_t(2V + y_t)$, of degree three instead of four. The resulting explicit polynomial is

$$\boxed{\begin{aligned} P_{22} = {} & \sum_{i=1}^{7} b_i^2 + \left((2r+1)^2c^4 - \Delta(f^2-1)\right)^2 \\ & + \left((V+y_t)^2 - \Delta(f^2-1)y_t(2V+y_t) - 1\right)^2 + b_{11}^2 = 0. \end{aligned}} \tag{9}$$

All abbreviations of Section 2 are retained except I, which no longer occurs, and y is replaced by y_t. There are 22 unrestricted integer variables and ten residuals.

Theorem 5.1. *The integer zero sets of P_{22} and P_{23} are in bijection under insertion/removal of $I = (2r+1)c^2$ and the stated root shear. The degree of P_{22} is exactly 14. It satisfies (1) with the same semantic dependencies as P_{23}.*

Proof. The row operation is invertible while $b_9 = 0$ is retained; the subsequent monic elimination and polynomial shear are uniquely reversible. A sum of integer squares is zero exactly when each summand is zero. The new strong residual has degree six. The centered auxiliary residual has degree seven, since $V = fo - c$ has degree two. All other residuals have degree at most six. The coefficient of $a^4 f^6 o^2 y_t^2$ in the final polynomial is exactly 4, arising uniquely from the square of the term $-2a^2 f^3 o y_t$ in the centered auxiliary residual. This proves the exact degree after specialization of all ports. □

The elementary row reduction is useful independently of this application: one should minimize degree *modulo the retained equations* before substituting high-degree definitions into every occurrence. Here direct substitution gives degree 20, row reduction gives 16, and the subsequent root centering gives 14, at the same 22 variables.

6 The sextic and octic companions

6.1 Twenty-six unrestricted integers at degree eight

Use the ordered variables

$$(J, F, \alpha, z, f, h, r, v, w, \tau, \eta, \zeta, y, \gamma, o, j, E, c, a, V, I, u, t, p_1, p_2, p_3).$$

Keep $q, C, H, \Delta, R, G, e, \mathrm{Pos}$ as before, but now set

$$X = 2u^2, \quad Y = (2v+1)q^3, \quad k = \eta + \zeta, \quad P = k + E(c - \eta), \quad D = X + ac + \gamma H.$$

The fourteen residuals are

$$\begin{aligned}
&X - 4qw, \quad a - E - Y, \quad EX + E - Xa, \quad c - (a - E)k - \eta,\\
&\tau^2 - P^2 + Pk - 1, \quad D^2 - \Delta c^2 - 1, \quad k - hE - R - 1,\\
&(K_0 + 4w)C - F - z(q-1), \quad V + c - fo, \quad V - cj - hE + k - 1,\\
&I - (2r+1)c^2, \quad I^2 - \Delta(f^2 - 1), \quad y^2 - I^2(y^2 - V^2) - 1,\\
&Ge - (t^2 - G - e + 1)^2 - \mathrm{Pos}(p_1, p_2, p_3).
\end{aligned}$$

Their sum of squares is P_{26}. Every residual has degree at most four. Only the last uses t, with coefficient -1 at t^4, so $[t^8]P_{26} = 1$. Keeping ζ independent is deliberate: it keeps $e = \eta\zeta$ quadratic and the joint guard quartic.

Soundness and canonical completeness use Section 3, followed by ten exact monic eliminations from the unswitched Q41 source. No signed-bootstrap theorem is needed for this companion. It has 26 unrestricted integer variables, exact degree eight, and the same consistency interpretation.

6.2 Thirty-one unrestricted integers at degree six

Start with the signed-bootstrap source and replace only the $F\alpha, FC$ guards by $G = \mathrm{Pos}(p_1, p_2, p_3)$. Retain the separate $\eta\zeta = \mathrm{Pos}(e_1, e_2, e_3)$ row. The variables are

$$(J, F, \alpha, f, h, r, v, \sigma, \eta, \zeta, y, \gamma, o, j, E, c, a, D, L, V, I, \mathcal{B}, \Delta, u, \\ e_1, e_2, e_3, Z, p_1, p_2, p_3).$$

Define $q = \beta J + 1$, $C = q - F - \alpha$, $k = \eta + \zeta$, $H = 4a + 3$, $X = D - ac - \gamma H$, $Y = (2v+1)q(L+1)$, and

$$R_L = ((q-1)(q-F) + \alpha)L + (M_C + qM_F)J.$$

The eighteen residuals are

$$\begin{aligned}
&L - q^2 + 1, \quad a - E - Y, \quad EX + E - Xa, \quad c - (a - E)k - \eta, \\
&\sigma^2 + (2\sigma - k)E(c - \eta) - 1, \quad \Delta - a^2 - 4a - 3, \\
&D^2 - \Delta c^2 - 1, \quad k - hE - R_L - 1, \quad (K_0 q + X)C - qF - Z(q-1), \\
&V + c - fo, \quad V - cj - hE + k - 1, \quad I - (2r+1)c^2, \\
&\mathcal{B} - I^2, \quad \mathcal{B} - \Delta(f^2 - 1), \quad y^2 - \mathcal{B}(y^2 - V^2) - 1, \\
&X - 4qu^2, \quad \eta\zeta - \mathrm{Pos}(e_1, e_2, e_3), \quad G - \mathrm{Pos}(p_1, p_2, p_3).
\end{aligned}$$

Here G is still the fixed determinant expression in q, F, C. Their sum of squares is P_{31}, of exact degree six: the residual $I - (2r+1)c^2$ contributes the unique coefficient 4 at r^2c^4. The centered coordinate is $\tau_{\mathrm{old}} = \sigma + E(c - \eta)$; this triangular substitution removes the potentially quartic term from the first norm before eliminating its product register. Six monic eliminations then leave the eighteen displayed rows.

7 Fixed coefficients and exact provenance

The integer family uses one coherent native tuple, not a mixture of small ports from different compilers. Its basic quantities are

$$\rho = 2^{121}, \qquad B = \rho^{8801} = 2^{1064921}, \qquad \beta = B - 1.$$

The shipped literal machine table has 430 transitions and starts in state 0. The deterministic coefficient constructor rebuilds the ordered 943 tile list from that table, compares every tile with the retained literal list, and computes the native ports by finite sums and shifts. It performs no unbounded search for a machine halt or a refutation.

For a compact exact description of those finite sums, let the ordered tiles be $T_i = (\ell_{i0}, \ell_{i1}, \ell_{i2}, \ell_{i3})$, $0 \le i < 943$. Set $w_j = 2^{10+2j}$. Define a coefficient map c_e by

$$\begin{aligned}
c_i &= 1 + \sum_{s=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(\ell_{is} + 512) w_{10s+j} + \sum_{j=40}^{43} w_j, \qquad 0 \le i < 943, \\
c_{o_s+2j} &= w_{10s+j}, \quad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \\
c_{1095} &= w_{40}, \quad c_{1096} = w_{41}, \quad c_{2195} = w_{42}, \quad c_{3296} = w_{43}, \quad c_{1063} = c_{1064} = 0.
\end{aligned}$$

Let $\mathcal{E}$ be these 989 support positions, including the two zero controller positions. Then

$$D_C = \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e\in\mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$

$$K_0 = D_C + B\rho, \qquad M_C = B - 1 - \sum_{e\in\mathcal{E}\setminus\{1\}} \rho^e - 2\rho^{1063},$$

$$\mu = 1022 + \sum_{j=0}^{43} w_j,$$

$$M_F = B - 1 + 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9}(\rho^{1000+2j} + \rho^{1021+2j}) + \rho^{2195} + \rho^{3296}$$
$$+ (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}.$$

All exponents in this fixed recipe are nonnegative. The finite tile list, the machine table generating it, and all bytes of the six resulting integers are included. They are alternative exact specifications of fixed integers; the hashes in the manifest are checks, not definitions of the coefficients.

The determinant alignment uses the leading coefficient $c_0 = 105534389824824194647382298625$ and the identities

$$121 \cdot 8791 + 1113 = 1064921 - 97, \qquad 121 \cdot 3296 + 666104 = 1064921 - 1.$$

Canonical native blocks have C's leading digit 1 at 3296 and F's leading coefficient c_0 at 8791. Together with $2^{96} \le c_0 < c_0 + 1 \le 3 \cdot 2^{95}$ and the source's carry bounds, these yield (6). The retained determinant proof gives the full finite-to-canonical argument.

Fixed integer	Bit length	Minimal unsigned bytes
β	1064921	133116
K_0	1065043	133131
M_C	1064921	133116
M_F	1064922	133116
A_*	1114	140
B_*	666105	83264
Total		615883

These coefficient costs are unchanged from package 71. The new savings are in variables and degree. The eleven-positive construction uses the distinct whole compact44 tuple with $\beta = 2^{590069} - 1$; it has its own binding and does not share coefficient leaves with the integer family. The nine-positive construction uses the distinct whole original31 tuple. Although its β is also $2^{1064921} - 1$, its other three ports differ from source34. Its raw four-port payload is 533099 bytes. Equality of the radix does not authorize mixing these tuples.

8 Eleven positive variables: degree 277

Package 70 combines three square roots into one norm and uses the strong Pell ordinate $I = iq(q-1)c^2$. The factors $q(q-1)$ ensure a coprimality condition used only to recover integrality of an auxiliary rational root. Package 69's following stronger lemma makes that condition unnecessary.

Lemma 8.1 (Rational Pell descent)**.** *Let $Q > 1$ and V be integers, and $y \in \mathbb{Q}$. If $QV^2-(Q-1)y^2 = 1$, then $y \in \mathbb{Z}$.*

Proof. Neither coordinate is zero; change signs to make them positive. Write $y = a/m$ in lowest terms, $m > 0$. The equation gives $m^2 \mid Q - 1$. If $V = 1$, then $y = 1$. For $V > 1$ define

$$V' = (2Q - 1)V - 2(Q - 1)y, \qquad y' = (2Q - 1)y - 2QV.$$

The first is integral, the norm is unchanged, and y' has the same reduced denominator m because $2Q - 1 \equiv 1 \pmod{m}$. The inequalities $y > V$ and $QV > (Q - 1)y$ follow from the original norm. They imply $-V < V' < V$, while the new norm excludes $V' = 0$. Replace the new coordinates by absolute values and induct on the positive integer V. The denominator is preserved until $V = 1$, proving $m = 1$. □

The new equation simply replaces I by ic^2 throughout package 70's literal norm. All unconditional radical separation and transport domination estimates need only $I \geq c^2$, which remains true for positive i. Once the norm recovers rational square roots, two are integral as rational roots of integer squares, and Lemma 8.1 makes the third integral. This is the only step at which the removed factor was used.

For an exact standalone polynomial specification, the positive unknowns are

$$(j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma).$$

With the compact44 coefficients, define

$$\begin{aligned}
&J = 3j + 1, \quad q = \beta J + 1, \quad C = q - F - \alpha, \quad E = XY,\\
&k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = EkY,\\
&R = ((q - 1)(q - F) + \alpha)(q^2 - 1) + (M_C + qM_F)J, \quad \widehat{R} = q^3(q + F + \alpha),\\
&M = q^3(q - 1), \quad B_t = (qK_0 + X)C - qF, \quad D_\tau = \ell(\ell + k) + 1,\\
&I = ic^2, \quad D_f = 1 + \Delta I^2, \quad Q = \Delta^2 I^2, \quad \delta = Q - 1,\\
&V = D_f(cT - R) - c, \quad D_n = QV^2 - 1, \quad \overline{V} = D_f(cT + \widehat{R}),\\
&L = 3\overline{V}, \quad W = 1 + (q - 1)L, \quad \overline{Z} = M(\ell + k + n + L) + (q - 1)Y - q^3 W B_t,\\
&r_0 = k - R - 1, \quad S = 4qY, \quad U = qS,\\
&D_* = U(X + ac) + H(U\gamma + SX - r_0), \quad A_1 = D_*^2 - U^2(1 + \Delta c^2),\\
&A = M^2 D_\tau, \quad B_f = M^2 D_f, \quad C_n = M^2 D_n,\\
&T_n = \delta^2 A B_f, \quad R_n = \delta C_n, \quad J_n = \delta(\overline{Z}^2 - A - B_f) - C_n,\\
&N_c = J_n^2 - 4\big(T_n + R_n(A + B_f)\big), \quad N_8 = N_c^2 - 64 T_n R_n \overline{Z}^2.
\end{aligned}$$

Then

$$\boxed{P_{11} = N_8 + A_1 + N_8 A_1 = 0.} \tag{10}$$

The full proof of both directions, including signed radical choices, positive reconstruction of the deleted transport gap, the product finalizer, and exact degree, is included in `positive11/PROOF.md`.

Briefly, the eight-root norm gives a rational signed sum of square roots of three positive, distinct radicands, with no zero signed sum. Hence all three roots are rational. The two integer radicands and Lemma 8.1 make every root integral. The unchanged bound $L > |f_*| + |y_*|$ restores the positive

transport gap $r = n + L - f_* - y_*$, and $|y_*| > |V|$ restores the positive auxiliary gap. In the completeness direction canonical Pell witnesses already permit $c^2 \mid I$, so the new parameter $i = I/c^2$ is positive. One chooses the weighted transport gap after the strong and auxiliary roots; its weight L is large enough for the final n to be positive.

The product finalizer is exact: $U = 4q^2Y$ gives $A_1 \equiv D_*^2 \pmod 4$, and the factor 64 gives $N_8 \equiv N_c^2 \pmod 4$. Thus neither residual is -2; no parity assumption on q is required. Thus $(1 + A_1)(1 + N_8) = 1$ forces both factors to be $+1$, and both residuals to vanish. Its exact degree is 277, certified by the nonzero coefficient

$$[j^{60}i^{24}\eta^{49}X^{42}Y^{85}\alpha^{16}\gamma]P_{11} = 2^7\,3^8(3\beta)^{60} > 0.$$

The literal division-free circuit uses 111 live binary additions, subtractions and multiplications, including scalar multiplications and squaring. Fixed integer leaves and register reuse are free in this metric. The two-residual system itself uses 108 operations. The degree reduction is $48/325 \approx 14.77\%$.

9 A direct majorant of the complete Pell radical

Another new construction improves a different measure of simplicity: the total degree with nine positive unknowns. It retains package 71's source encoding and replaces its final square-root bound. The resulting polynomial has degree 1 158 228, compared with 1 379 730 for that package, and its explicit circuit has 1 753 arithmetic operations, compared with 478. Thus this is a degree–circuit-size tradeoff. The improvement is approximately 16.05% in degree. It neither lowers the number of unknowns nor asserts a global minimum.

9.1 Why approximate the complete radical?

The previous finalizer constructs a polynomial upper bound for $\sqrt{DF}\,G$ and then multiplies it by H. Errors in that bound are therefore multiplied by a high-degree polynomial. We instead construct a polynomial upper bound for the complete radical

$$\sqrt{DF\big((G^2-1)H^2+1\big)}.$$

The discovery calculation uses a formal expansion at $A = \infty$. The proof of the resulting bound consists entirely of finite integer polynomial identities and positive coefficients.

In this subsection A, C, i, Z are independent variables. Define

$$\begin{aligned}
J &= 4A - 5, & D &= (A^2-1)C^2 + 1,\\
e &= iJC, & E &= eDC,\\
F &= (A^2-1)E^2 + 1, & L_0 &= CD + 1 - A,\\
G &= A + FL_0, & H &= C + FZ,\\
I &= (G^2-1)H^2 + 1, & Q_* &= DFI.
\end{aligned}$$

As a polynomial in A, Q_* has degree 46 and leading coefficient

$$(1024C^{24}i^5Z)^2.$$

Write $Q_* = \sum_{k=0}^{46} q_k A^k$. Define $t_{23}, \dots, t_0$ in $\mathbb{Q}(C, i, Z)$ by the finite recurrence

$$t_{23} = 1024 C^{24} i^5 Z,$$
$$t_{23-k} = \frac{q_{46-k} - \sum_{\ell=1}^{k-1} t_{23-\ell} t_{23-k+\ell}}{2 t_{23}} \qquad (1 \le k \le 23). \tag{11}$$

This is the polynomial part of the formal square root with positive leading term. The recurrence is a finite specification of coefficients; no division by an unknown remains in the final equation. Set

$$\mathcal{P}(A, C, i, Z) = 65536 C^7 \sum_{r=0}^{23} t_r A^r, \qquad \sigma = 65536 C^7. \tag{12}$$

Exact coefficient arithmetic proves $\mathcal{P} \in \mathbb{Z}[A, C, i, Z]$. It has 654 nonzero terms, all of which are supplied in `nine_positive/data/radical.json`. The same file gives every coefficient of

$$\mathcal{D} = \mathcal{P}^2 - \sigma^2 Q_*. \tag{13}$$

The coordinate degrees of these two polynomials are

$$\deg_{(A,C,i,Z)} \mathcal{P} = (23, 31, 5, 1), \qquad \deg_{(A,C,i,Z)} \mathcal{D} = (22, 42, 10, 2).$$

There are 3781 nonzero defect terms. Denominators in the formal coefficient recurrence cancel after the displayed factor $65536 C^7$; there are no negative exponents in either delivered table.

9.2 A uniform inequality certified by positive coefficients

The decisive certificates are the following exact identities, represented by their complete coefficient lists:

1. $\mathcal{P}(X + 3, Y + 1, i, Z)$ has 1816 nonzero coefficients, all strictly positive.
2. $\mathcal{D}(X + 3, Y + 1, U + 1, V + 1)$ has 32 637 nonzero coefficients, all strictly positive. Its constant coefficient is
$$295870595720775501924817.$$

The lists are in `certificates/P_shifted.json` and `certificates/defect_shifted.json` within the nine-positive component. Regeneration by rational coefficient arithmetic and an independent implementation using only integer sparse polynomials agree. These are finite algebraic certificates for statements over the whole domain, rather than sampled inequalities.

Lemma 9.1 (Complete-radical majorant)**.** *For all integers $A \ge 3$ and $C, i, Z \ge 1$,*

$$\mathcal{P}(A, C, i, Z) > 65536 C^7 \sqrt{DFI} > 0. \tag{14}$$

Proof. Here J, D, E, F are positive. Since $CD + 1 - A \ge A^2 - A + 1 > 0$, we have $G > A \ge 3$; also $H, I > 0$. The first certificate evaluated at $X = A - 3$, $Y = C - 1$ gives $\mathcal{P} > 0$. All its monomials are nonnegative, and its terms of X- and Y-degree zero are positive when $i, Z \ge 1$. The second certificate evaluated at $(X, Y, U, V) = (A - 3, C - 1, i - 1, Z - 1)$ gives $\mathcal{D} > 0$. Strictness holds even on every boundary because its constant is positive. Equation (13), together with $\mathcal{P} > 0$, now gives (14). □

The use of a formal expansion in the discovery process therefore does not introduce an asymptotic qualification into the theorem. The bound holds before any solution of the final equation has been assumed.

9.3 The fixed equation and its nine unknowns

All nine unknowns

$$b, g, h, s, w, C, i, j, n \in \mathbb{Z}_{>0}$$

retain the domain and meaning in the copied source-71 prefix. The source polynomials R, N, t are the fixed source-65 encoding polynomials; their complete weights, target positions and coefficient kernels are included in the component's data directory. The whole original-31 native coefficient tuple is supplied by literal payloads and a finite generator. No coefficient is left as a free program parameter.

For clarity, the ratio registers needed below are

$$\begin{aligned}
U &= N^2 w, & Y &= N^2 s, & M &= RY,\\
P_0 &= 2M^2 U, & K &= R + 1 + h(P_0 - 1), & A &= M(U + 1),\\
B_1 &= 2R + 1, & W_1 &= bw, & J &= 4A - 5,\\
Q_0 &= 3W_1 C - 2(W_1^2 - 1), & & & &\\
m &= K^2(4t - g) - 16t(C - KY)^2, & & & &\\
\Omega &= Jm, & E_0 &= J(n + W_1^2) + Q_0, & V_0 &= P_0 K,\\
a_0 &= (P_0^2 - 1)K^2 + 1, & Z &= B_1 + (j - 1)C. & &
\end{aligned}$$

In the local polynomials above, substitute this A, this C, the free i, and this Z. In particular $H = C + FZ$ is exactly the inherited Lucas-certificate register. Define

$$V_1 = \mathcal{P}(A, C, i, Z), \qquad a_1 = (65536C^7)^2 DFI, \qquad \Theta = E_0 - \Omega(V_0 + V_1). \tag{15}$$

The new equation is

$$\boxed{F_9 = (\Theta^2 - \Omega^2(a_0 + a_1))^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{16}$$

Every named register is a polynomial abbreviation, not an additional unknown. The division-free DAG `equation9.json` is the complete literal instance. The source bounds $R, N \geq 8$ imply $A \geq 3$ on every ambient positive input, and $C, i, Z \geq 1$ is immediate. Consequently the majorant lemma applies without a circular appeal to the final equation.

9.4 Why the replacement preserves the ratio theorem

The original first majorant satisfies $V_0 > \sqrt{a_0} > 0$, since $V_0^2 - a_0 = K^2 - 1 > 0$. The lemma gives $V_1 > \sqrt{a_1} > 0$. Moreover

$$E_0 = Jn + (J - 2)W_1^2 + 3W_1 C + 2 > 0.$$

Thus a zero of (16) cannot have $m = 0$: that would give $F_9 = E_0^4 > 0$. Factoring the two-root norm over the reals gives signed roots $r_0^2 = a_0$, $r_1^2 = a_1$ with

$$\frac{E_0}{J} = m(V_0 + V_1 + r_0 + r_1). \tag{17}$$

The strict majorants make the parenthesis positive for every sign choice, so $m > 0$ follows before any root-integrality argument. The unchanged margin argument now recovers the required free-carrier bounds and makes $DFI > a_0$, hence $a_1 > a_0$.

The sum $r_0 + r_1 = \Theta/\Omega$ is rational and nonzero. The identity

$$2(r_0 + r_1)r_0 = (r_0 + r_1)^2 + a_0 - a_1$$

therefore makes both roots rational, and an integer with a rational square root has an integral square root. Although the scale $65536C^7$ depends on the current positive input, it is a nonzero integer. Consequently $r_1/(65536C^7)$ is a rational square root of DFI and is also integral. No fixed-scale assumption is needed. Equation (17) then recovers $J \mid Q_0$ exactly as before. All the inherited ratio conditions, and hence the source zero, follow.

Conversely, a source zero supplies the original ratio witnesses and integral roots. Choose the positive roots $r_0 = \sqrt{a_0}$ and $r_1 = 65536C^7\sqrt{DFI}$, and take

$$n = m(V_0 + V_1 + r_0 + r_1) - W_1^2 - Q_0/J.$$

This is an integer. The unchanged estimate $W_1^2 + Q_0/J < 2D$, together with $F > D$, $I \geq 1$ and $V_1 > 65536\sqrt{DFI} > 2D$, shows $n > 0$. One norm factor vanishes. This proves the same source-zero equivalence as source 71. Composing with the retained native-source and syntactic refutation calibration, and negating, gives the ordinary theorem formalizable in ZFC:

$$\mathsf{ZFC} \vdash (F_9 \text{ has no positive integer zero}) \iff \mathrm{Con}(\mathsf{ZFC}).$$

The new component supplies no additional end-to-end Lean or object-language ZFC certificate; the inherited calibration boundary is stated explicitly in its proof and provenance files.

9.5 Exact degree and the change of dominant term

Let $r = 7963$, $\nu = 2115$, $W = 116$ be the degrees of R, N and the maximum source weight, and put $a = r + 4\nu + 2 = 16425$. Then

$$\deg \Omega = 5a + 2W + 3 = 82360, \qquad \deg V_0 = 3a + r + 1 = 57239,$$
$$\deg \delta_0 = 3a + r + 2 = 57240, \qquad \delta_0 = V_0^2 - a_0 = K^2 - 1.$$

The unique highest weighted-degree term of $\mathcal{P}$ is $2^{26}A^{23}C^{31}i^5Z$, giving

$$\deg V_1 = 23a + r + 36 = 385774.$$

The corresponding unique term of its defect is

$$141664530202624\, A^{22}C^{42}i^{10}Z^2,$$

so $\delta_1 = V_1^2 - a_1$ has degree $22a + 2r + 52 = 377328$. To resolve the cancellations in the outer norm, set

$$B_* = E_0^2 - 2E_0\Omega(V_0 + V_1) + \Omega^2(\delta_0 + \delta_1).$$

It has degree 542048, and the exact identity

$$F_9 = B_*^2 + 4\Omega^2V_0V_1B_* + 4\Omega^4(\delta_0V_1^2 + a_0\delta_1) \tag{18}$$

shows the following four degrees:

B_*^2	1 084 096
$4\Omega^2 V_0 V_1 B_*$	1 149 781
$4\Omega^4 \delta_0 V_1^2$	1 158 228
$4\Omega^4 a_0 \delta_1$	821 246

The third term is uniquely largest. Its leading coefficient is positive, so the exact degree is

$$\boxed{\deg F_9 = 72r + 276\nu + 8W + 224 = 1\,158\,228.}$$

The dominant term is different from that of the previous finalizer; merely substituting the new defect degree into its old formula would be incorrect. If ρ, σ_N, τ are the positive diagonal leading coefficients of R, N, t, respectively, the new leading coefficient is $2^{90}\tau^4\rho^{72}\sigma_N^{276} > 0$. For the actual source constants $\alpha = 2^{2134939}$ and $c = 2^{17079501}$ this becomes

$$2^{162}\alpha^{578828}c^{276} > 0.$$

Thus specialization to the actual native coefficients cannot remove the leading term.

The final circuit contains 923 multiplications, 810 additions and 20 subtractions. This count includes every multiplication by a fixed integer and counts reuse once. The independent verifier reconstructs all 654 majorant coefficients directly from the final DAG, checks the whole finalizer as a polynomial identity, and verifies 45 retained named source expressions against source 71. Both independent checkers use only the Python standard library and pass with and without Python's optimization flag.

10 Proof boundary and reproducibility

The new transformations are exact ordinary mathematical results. The semantic dependency is the ordinary native/Pell calibration presented in the cited repository manuscripts, including all-input soundness and canonical completeness for the exact coefficient tuple. The present work proves new reductions using that theorem; it does not independently close the repository's remaining formal native-calibration or internal-ZFC obligations. The unbounded semantic input states that the specified source-34, compact44, or original31 system has a solution if and only if the fixed calculus has a finite ZFC refutation, with the documented canonical completeness witnesses. The proofs above use its all-input soundness and its stronger canonical completeness, with the literal coefficient bindings. They do not infer either from finite simulations.

Every new integer zero first determines the old product guards and signed arithmetic source, which in turn determines a finite ZFC refutation. In the other direction one regenerates a canonical source witness, preserves its determinant estimates, supplies a nearest-square guard, and applies the exact coordinate maps. These ordinary arithmetic arguments can be formalized in ZFC. Both the full formal native/compiler calibration and a machine-checked proof of the displayed object-language sentence are separate deliverables, and are not supplied here.

In particular the repository's `OBJECT_ZFC.md` correctly distinguishes an external Lean equivalence from a proof in the sealed ZFC calculus. Truth on external natural numbers, numeralwise agreement, and bounded consistency proofs alone do not settle nonstandard internal inputs. Nothing in this report treats those weaker statements as the requested internal biconditional. The local polynomial row reductions, monic maps, and sign lemmas can be inserted directly into that formalization once its formal native or direct-checker endpoint is complete. This is a distinction between ordinary theorem dependencies and their formal verification, not an inference from finite replay to an unbounded consistency statement.

Executable evidence. The archive includes literal expanded residual packets, compact formulas, exact coefficient payloads and regeneration code, source pins and licences, independent algebraic checks, a top-level verifier, and this LaTeX source. The integer checker expands over symbolic fixed ports, verifies every elimination and parent restoration, and exhibits coefficients that remain nonzero on the actual fixed slice. The independently written audit checks the final formulas without importing the construction code. Normal and optimized Python executions retain all explicit checks; finite controls are labelled as controls, not as proofs of unbounded claims.

Useful next formalization tasks. The nearest-square lemma, determinant implication, canonical-interval composition, and elementary norm-row operation are small reusable targets. The last has an especially simple ring certificate. These can improve the closed candidate while the larger native-calibration and internal-checker projects proceed. The new low-degree forms do not depend on Jones or Sun compression; their inherited source still uses the K215 native encoding.

A An alternative route: one loop for unordered proof certificates

This section gives a source-level improvement independent of Turing-machine simulation and Jones compression. It does not claim a new measured final polynomial. The affine-product construction in package 68 uses one bounded universal block for proof steps and a second for a common-multiple condition. Padding the proof makes the two blocks share one index. The resulting source has four outer natural parameters instead of five and one bounded universal block instead of two. The three separate label bounds are merged into one joint bound, reducing its local witness count from 23 to 22. The number of witnesses after arithmetic closure has not been evaluated.

A.1 Statement

Let a Horn calculus have natural-number labels, a freely derivable truth label 0, and rules $(p, q) \longrightarrow c$ with one conclusion and two premise slots. Axioms have premises $(0, 0)$; unary rules are padded by 0. Include the harmless rule $(0, 0) \longrightarrow 0$ and fix target 1. Write D for its least set of derivable labels.

For $M \geq 1$, put $f_M(j) = 1 + jM$. Let

$$\beta(x, y, i) = x \bmod (1 + (i + 1)y).$$

The following predicate has outer natural parameters T, x, y, M_0, where $M = M_0 + 1$:

$$\beta(x, y, 0) = M + 1, \qquad \beta(x, y, T) = 1, \tag{19}$$

$$\forall i < T \quad \exists p, q, c, d, k \in \mathbb{N} : \quad (p, q) \longrightarrow c,$$

$$p + q + c + d = T, \tag{20}$$

$$M = (i + 1)k, \tag{21}$$

$$\beta(x, y, i + 1) f_M(c) = \beta(x, y, i) f_M(p) f_M(q). \tag{22}$$

The rule relation in this display can have its own fixed local witnesses. Only one occurrence of bounded universal quantification is present.

Theorem. The predicate has a solution if and only if $1 \in D$. This equivalence is provable by elementary finite mathematics in ZFC.

A.2 Soundness

The endpoints imply $T > 0$: at $T = 0$ the same beta value would equal both $M + 1 \geq 2$ and 1. Equation (21), as i runs from 0 to $T - 1$, says that every positive integer at most T divides M. For distinct $1 \leq a < b \leq T$, the weights $f_M(a)$ and $f_M(b)$ are coprime. Indeed, a common divisor d divides $(b - a)M$ and is coprime to M, hence divides $b - a$. But $b - a \mid M$, forcing $d = 1$.

Put $S_i = \beta(x, y, i)$. Since all the weights are positive, $S_0 = M + 1 > 0$ and (22) imply $S_i > 0$ for every $i \leq T$. Multiplying the transition identities and cancelling the positive intermediate states gives

$$\prod_{i<T} f_M(c_i) = f_M(1) \prod_{i<T} f_M(p_i) f_M(q_i).$$

Pairwise coprimality of the positive-label weights, together with $f_M(0) = 1$, forces exact equality of positive-label multiplicities. For example, choose one prime divisor of each $f_M(j)$ and compare its valuation; that divisor occurs in no other weight. In the free abelian group on positive labels,

$$\sum_{i<T} (e_{c_i} - e_{p_i} - e_{q_i}) = e_1, \qquad e_0 := 0.$$

If $1 \notin D$, sum the finitely many coordinates outside D. A rule with conclusion outside D has at least one premise outside D, while a rule with conclusion inside D contributes at most zero. Thus the left side has nonpositive sum and the right side has sum 1, a contradiction.

This argument does not accept cyclic proofs. Its use of one conclusion per rule is essential; the corresponding assertion for several conclusions is false.

A.3 Completeness and the order of the choices

Take a finite proof tree of label 1. Expand it from its root, recording a rule every time an open nonzero obligation is replaced by its two premises; zero premises are discarded. Let L be the number of recorded rules and m the greatest sum $p + q + c$ in a recorded rule. Choose

$$T \geq \max(1, L, m), \qquad M = T!, \qquad M_0 = M - 1.$$

Pad the list to length T with $(0, 0) \longrightarrow 0$. These padding steps leave the accumulator unchanged because $f_M(0) = 1$.

At each stage let S_i be the product of $f_M(j)$ over the current multiset of nonzero open obligations. Removing the chosen conclusion and inserting its premises proves (22). The initial value is $M + 1$; the final value is 1. Every row-label sum is at most T, supplying the joint slack; and the quotients $k_i = M/(i + 1)$ establish (21).

Only after these finite integers have been chosen do we choose a beta scale. For instance, take

$$y = T!(1 + \max_{0 \leq i \leq T} S_i).$$

Then the moduli $1 + (i + 1)y$, for $0 \leq i \leq T$, are pairwise coprime: their index differences are at most T and divide y. They also exceed the required residues. The finite Chinese remainder theorem supplies x with $\beta(x, y, i) = S_i$ simultaneously. This completes the proof. The scale y and the common multiple M are distinct witnesses; identifying them would generally destroy completeness. The arithmetic closure compiler's own scales must likewise remain independent.

A.4 Literal local-polynomial adapter

Package 68 supplies a nonnegative local polynomial $H(S, S', m, M_0; \mathbf{u})$ with 23 natural witnesses and degree 50. It checks one padded numeric C222 rule, its three label bounds, and its affine accumulator update. Write H_0 for its Horn product plus accumulator-update square, removing the three old bound squares and their slack coordinates. Here $\mathbf{u}$ denotes the twenty retained witnesses and includes the three label coordinates p, q, v. With a fresh joint-bound witness b_0, define

$$J(S, S', T, M_0, i; \mathbf{u}, b_0, k) = H_0(S, S', M_0; \mathbf{u}) + (p + q + v + b_0 - T)^2 + \big(M_0 + 1 - (i + 1)k\big)^2.$$

The retained nonnegativity makes $J = 0$ precisely the conjunction of the rule/update, joint-bound, and divisor conditions. This source has five exposed step coordinates and 22 local natural witnesses. Its degree remains exactly 50: the new summands have degrees at most 4 and cannot remove the degree-50 part of the Horn product.

One now applies the supplied bounded-forall constructor once to the single family $J = 0$, conjoins the beta endpoint conditions, and existentially binds the four outer parameters. This is a precise source recipe, but the full ordinary polynomial and its total cost have not been materialized here. The old C222 calibration remains an inherited obligation if this source is used as a Con(ZFC) endpoint. No result in this section fills the repository's separate gap between an external theorem and a checked internal ZFC proof.

The executable controls test 988 artificial Horn systems with up to three rules on labels $0, 1, 2$. They construct and verify a certificate for all 185 systems in that family where target 1 is derivable, including beta encoding, and reject all 185 corrupted divisor quotients. A separate counterfeit uses $M = 1$, axiom 3, and rule $(1, 0) \longrightarrow 0$: the accumulator $2, 4, 1, 1$ passes the raw update but fails the divisibility condition. These tests support implementation checks; the unbounded theorem is the proof above.

B Sources and attribution

The archive records exact source paths and hashes in `PROVENANCE.json`. The following references identify the mathematical dependencies, rather than merely similar constructions.

1. ProveIt, pinned wave-ten review, https://github.com/VladimirReshetnikov/ProveIt/blob/2605669a532342f153b13eb6b4a0af4b54f5fa37/Computability/DiophantineConsistency/PROPOSAL_REVIEW_WAVE10.md.

2. Package 34, `coefficients/build_coefficients.py`, `COEFFICIENT_PROOF.md`, and source-34 integer/Pell calibration.

3. Package 31, the original mixed-radix coefficient tuple and its native calibration; package 44, the distinct compact tuple used for eleven positives.

4. Package 64, `integer41/PROOF41.md`, `PROOF44.md`, and the literal `quartic41.json`.

5. Package 68, `integer_system/PAIR_GUARD_PROOF.md`, and its direct affine-product proof-certificate construction.

6. Package 69, `positive/PROOF.md`: rational Pell descent.

7. Package 70, `eleven/PROOF.md` and `positive/section.tex`: three-root transfer and its weighted positive transport interface.

8. Package 71, `proofs/DETERMINANT_GUARD.md` and the finite source-34 coefficient binding.

9. Package 65, the fixed nine-positive encoding prefix; package 59, the inherited ratio theorem; package 71, the comparison nine-positive norm and majorant. Their literal source files are retained in the nine-positive component.

10. Package 75, `guards/PROOF.md` and `guards/review/MATHEMATICAL_REVIEW.md`: signed-power bootstrap.

11. ProveIt, `OBJECT_ZFC.md` at the same full commit: the current distinction between external equivalence and an internal ZFC derivation.

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